

# SkillClaw

Collective Skill Evolution for Multi-User LLM Agent  
Ecosystems

[arxiv.org/abs/2604.08377](https://arxiv.org/abs/2604.08377)

# Agent Skills Remain Static After Deployment

Users Rediscover Solutions Independently

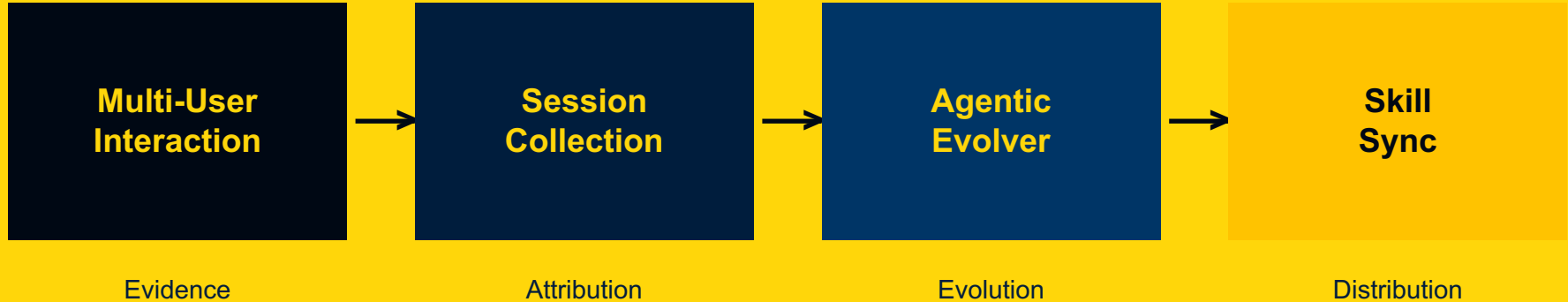
## Current Reality

- Skills manually installed from centralized hub
- Solutions from interactions don't persist
- Same failures observed repeatedly across users
- No system-level knowledge accumulation

## The Gap

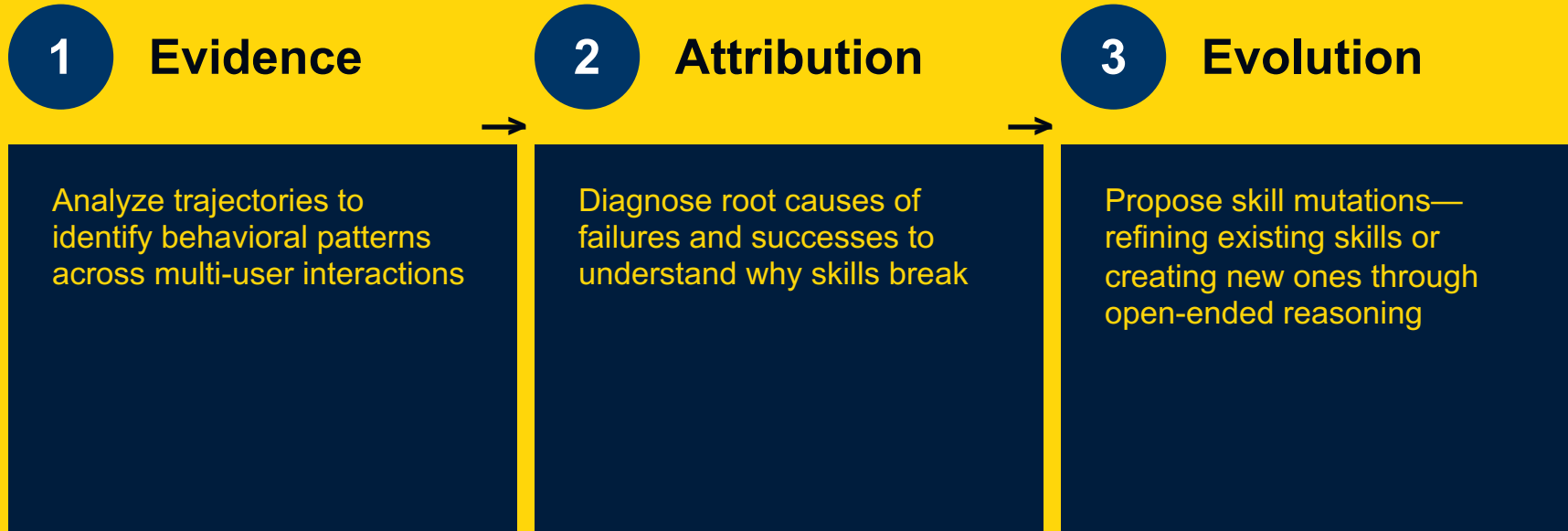
- Memory methods tied to specific instances
- Skill libraries treated as static resources
- No evolution through usage

# Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution



The agentic evolver performs open-ended reasoning over interaction evidence—not predefined update rules—to propose skill mutations verified before merging into the shared repository.

# Agentic Evolver: Evidence → Attribution → Evolution



Open-ended agent reasoning—not predefined rules—drives skill evolution

# Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

## Why Cross-User Aggregation?

- Different users exercising the same skill under diverse contexts produce complementary views
- Reveals conditions under which a skill works AND when it breaks
- Single user rarely generates enough signal to separate generalizable improvement from idiosyncratic fix

## Compatible Claw-Style Systems

OpenClaw

CoPaw

IronClaw

PicoClaw

ZeroClaw

NanoClaw

NemoClaw

Aggregated evidence from diverse user interactions enables the evolver to distinguish generalizable improvements from one-off fixes.

# Case Study: Multimodal Creative Pipeline — 6 Days of Evolution

| Day | Candidate Skill                             | Outcome | Rationale                  |
|-----|---------------------------------------------|---------|----------------------------|
| 1   | validate-tmp-workspace-inputs               | Accept  | Upgrade to new best pool   |
| 2   | multimodal-input-validation-and-setup       | Reject  | Keep current best pool     |
| 3   | multimodal-creative-task-pipeline           | Reject  | Keep current best pool     |
| 4   | multimodal-creative-task-pipeline (impr.)   | Reject  | Keep current best pool     |
| 5   | ...pipeline + validate-required-input-files | Reject  | Keep current best pool     |
| 6   | multimodal-creative-task-pipeline (cand.)   | Reject  | Not admitted to next cycle |

**1 of 6**  
accepted

Conservative verification prevents regressions — only skills that demonstrably improve behavior are accepted.

# Case Study: Git Push Auth-Fallback — Iterative Refinement Over 6 Days

| Day | Candidate Skill                                      | Outcome | Rationale                                                       |
|-----|------------------------------------------------------|---------|-----------------------------------------------------------------|
| 1   | git-push-with-auth-fallback                          | Accept  | Safe fallback instead of blocking on push failure               |
| 2   | git-push-with-auth-fallback                          | Accept  | Unified patch/bundle filenames, reduced inconsistency           |
| 3   | git-push-with-auth-fallback + git-clone-to-directory | Accept  | Auth-alternative paths & secrets audit; fixed subshell pitfalls |
| 4   | (none — same pool retest)                            | Accept  | Validator confirmed current pool as best                        |
| 5   | git-push-with-auth-fallback                          | Reject  | Added push hang handling; no improvement                        |
| 6   | git-push-with-auth-fallback                          | Reject  | Added identity config; no improvement — plateau reached         |

3 of 6

accepted

Successful iterative refinement: evidence-supported improvements accepted until plateau reached.

# Key Takeaways

## 1 Collective Evolution

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem

## 2 Fully Automatic

Skill evolution driven by runtime interaction without manual curation or explicit user intervention

## 3 Agentic Evolution Paradigm

Skill updates produced through open-ended reasoning rather than predefined update rules

Implications: Building continuously improving agent systems where skills evolve from real-world usage, not manual curation